RESAC 2026 · MEMORIAL UNIVERSITY · PANEL 4

Digitizing Ancient Religious Texts in the Age of AI

From Author to Editor: Moving the Scholarly Team Up One Rung

Dr. Ken M. Penner & Demarquis Moss · St. Francis Xavier University · SSHRC IDG 430-2026-01662

GREEK UNCIAL TRANSCRIPTION

4–5 Hours → ~20 Min

- Manual transcription: 4–5 hours per page of 6th-c. uncial scriptio continua.
- Auditing AI draft: ~20 minutes for rigorous human verification.

→ Order-of-magnitude reduction in keystroke labor.

LEMMAS & MORPHOLOGICAL TAGGING

Days of Work → ~5 Min

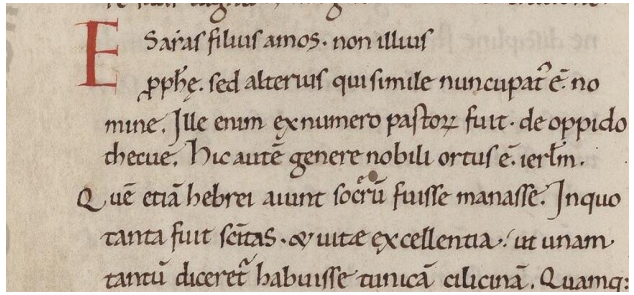
- Manual tagging: Days of inflectional parsing and lexical lookup.
- AI-assisted pass: ~5 minutes of automated machine inference.

→ Human checking is 5–10× faster than tagging from scratch.

SLIDE 2 · EXPERIMENTAL DESIGN

What We Tested: editio-bench

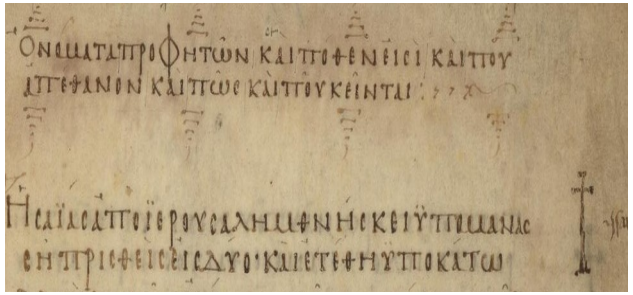
One benchmark, three manuscripts across a damage gradient, and the entire editorial workflow



EASY / CLEAN

Latin: Vat.lat.629 f. 3v

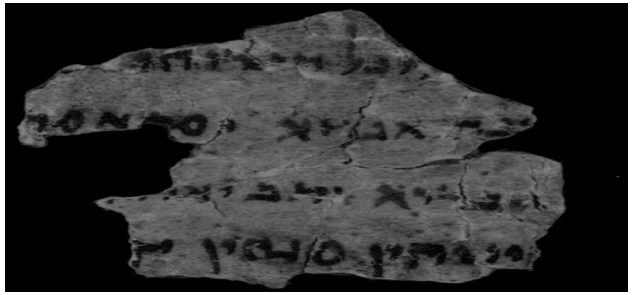
Caroline minuscule (12th c.)



MODERATE / UNCIAL

Greek: Vat. gr. 2125 p. 11

Codex Marchalianus (6th c.)



VERY HARD / DAMAGED

Aramaic: 4Q530 Frag. 6

Book of Giants (DSS leather)

Transcription → Collation → Translation → Lacunae → Morphology → TEI XML

13 Scored Tasks · 44 Models · 57 Verified Configurations · Three-Judge Consensus

SLIDE 3 · WORKFLOW ARCHITECTURE

The Escalation Cascade: 90 / 10 / 1

The saving comes from a scholarly filter, not from full autonomous generation

100% — COMPUTER DRAFTS EVERYTHING

Machine generates candidate text across all tokens

~10% — RESEARCH ASSISTANT REVISES

R.A. audits, catches errors, and corrects the draft

~1% — ESCALATED TO SUPERVISOR / P.I.

Only genuinely difficult Cruces reach senior review

EMPIRICAL GROUNDING

Greek Codex Marchalianus p. 11

- **Best model CER: 0.0365 (< 4% Error)**
- ~96% of characters right before human review.
- The R.A.'s 20-minute check IS the 10% filter.

→ Effort compresses because scholars edit rather than authoring from scratch.

** 90/10/1 is our project working impression, consistent with benchmark error distributions.*

SLIDE 4 · EPIGRAPHIC LIMITS

Limitations: Where the Cascade Breaks

The cascade holds on clean manuscripts, but breaks completely on degraded physical materials

MANUSCRIPT TIER	MEDIAN CER	CONFIGS ≤ 0.05	SCHOLAR'S ROLE
Latin: Vat.lat.629 (Caroline minuscule)	0.03 (97% correct)	37 of 57 configs	**Editor (Cascade Holds)**
Greek: Vat. gr. 2125 (Uncial scriptio continua)	0.13 (best 0.0365)	14 of 56 configs	**Editor (Cascade Holds)**
Aramaic: 4Q530 (Broken leather fragment)	0.80 (best 0.63)	0 of 57 configs	**Author (Cascade Breaks)**

ARAMAIC VISUAL SCORE: 7.0 / 100

- Text-based tasks score 72–87 / 100.
- Visual transcription of damaged leather collapses to 7.0 / 100 average.
- Frontier model gets 63% of letters wrong.

THE CARDINAL EPIGRAPHIC RULE

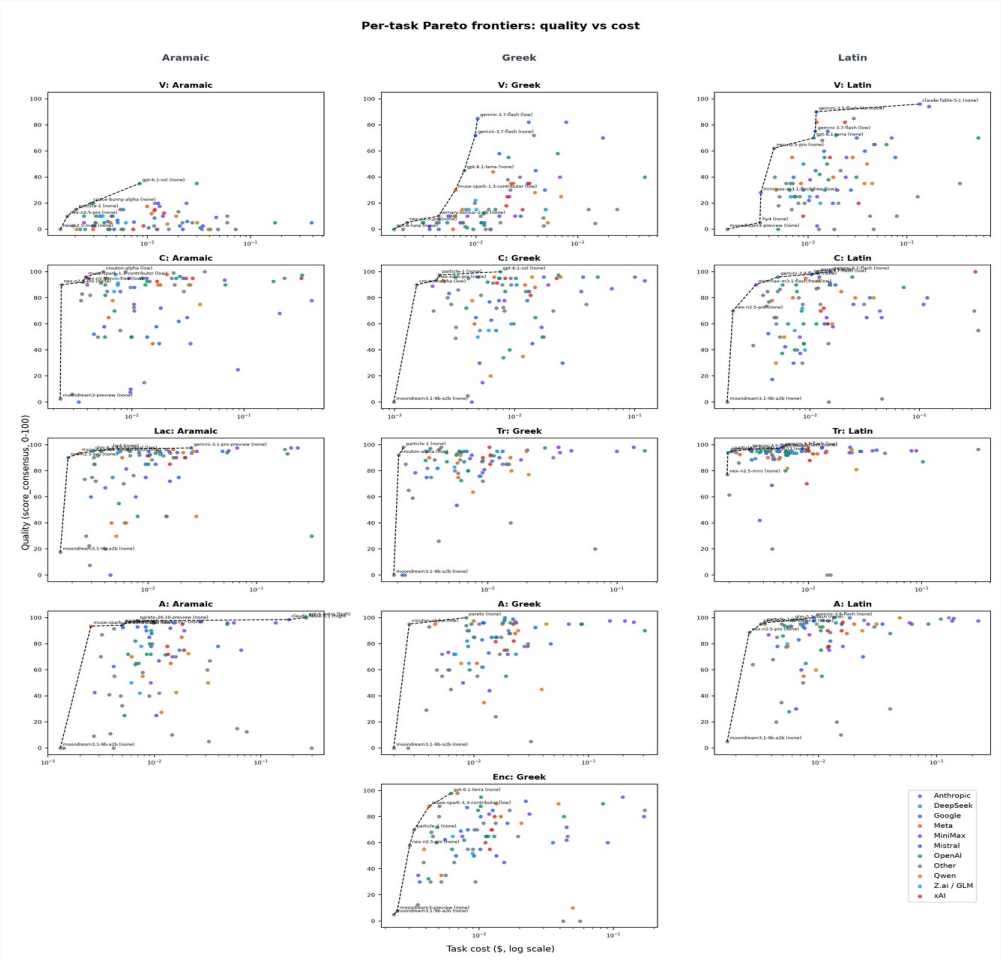
- On legible manuscripts: The AI draft is high quality; the scholar acts as an Editor.
- **On degraded fragments: There is nothing worth editing; the scholar is still the Author.**

SLIDE 5 · MODEL SELECTION STRATEGY

Use Different Models for Different Tasks and Languages

No single model dominates; effective projects assemble task specialists

TASK / CORPUS	LEADING MODEL (SCORE / 100)
Collation	Grok 4.7 (96.7) · GPT-6 Astra (94.5)
Translation	Gemini 3.1 Pro (97.0) · Grok 4.7 (97.0)
Latin Corpus	Gemini 3.7 Flash (92.4)
Greek Corpus	Gemini 3.1 Pro (91.7)
Aramaic Corpus	Claude Fable 5.1 (87–91) · Next: 78.5



BUDGET SPECIALISTS VS. SINGLE BEST

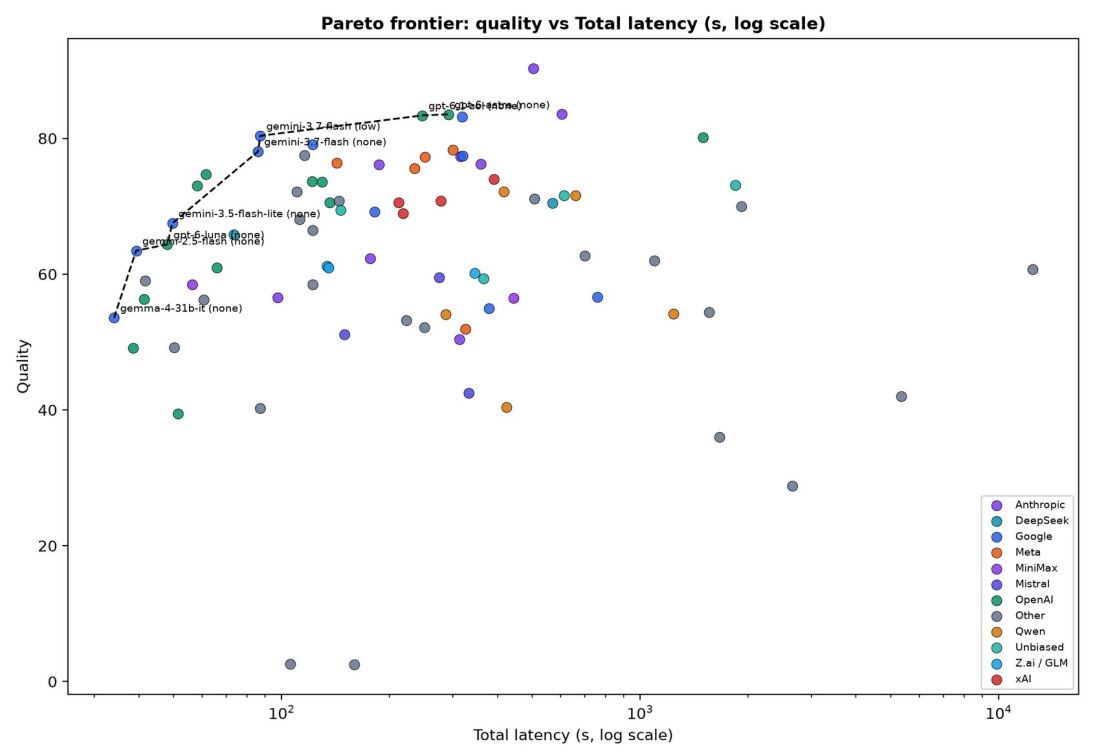
- Best model per task (< $0.20 / suite) ➔ 89.2 Score
- Single best model overall ➔ 90.3 Score at $1.72

➔ Save 8× budget for ~1 point of difference.

SLIDE 6 · LATENCY & EFFICIENCY

Speed and Cost Tend to Correlate

Fast is usually cheap, and waiting longer does not buy higher philological quality



Cost vs. Latency: r = 0.57 | Quality vs. Latency: r ≈ 0.04

MODEL	NONE	LOW	HIGH	EXTRA
Claude Fable 5.1	90.3	—	**83.6**	+22%
GPT-6 Astra	83.5	—	**80.2**	4.4×
Gemini 3.7 Flash	78.1	80.4	**77.5**	2×

→ Practical Rule: Set reasoning effort low or off. Extended thinking loops do not improve ancient text decipherment.

SLIDE 7 · SCHOLARLY ALLOCATION

Where Human Effort Goes: The Shaping Stages

AI compresses mechanical drafting; human effort concentrates at the two ends

THE BOW-TIE MODEL OF EDITORIAL LABOR

- BEFORE AI: Effort bulged in the middle (months of manual transcription, collation, tagging).
- **WITH AI: The middle is compressed. Human discernment concentrates at:**
  - **➔ INPUT END: Setting parameters, standards, baselines, and philological goals.**
  - **➔ OUTPUT END: Auditing results, resolving Cruces, and critical interpretation.**

INPUT: STANDARDS DECISION

Greek Ground Truth & Diplomatic Baseline

- Initial CER was ~9.5% because ground truth tested normalized text.
- Ken's ruling on diplomatic baseline (State C) dropped CER to 0.0365 (<4%).

➔ Apparent difficulty was manufactured by human standards.

OUTPUT: HUMAN AUDIT CATCHES

Catches That Automated Scripts Missed

- Caught a headline 70.5 score resting on only 2 of 13 tasks.
- Unmasked duplicate cache replays across runs.
- Caught AI judges giving 100/100 on empty images.

➔ Discovered by human auditors, not automated scripts.

SLIDE 8 · TEAM ROLES

Everyone Moves Up a Rung

At every level, the human role shifts from author to editor

THE ELEVATION OF RESEARCH ROLES

→ P.I. / Senior Scholar
Higher-level design, theoretical synthesis, ethics

→ Supervisor / Postdoc
Methodological direction (what the P.I. used to do)

→ Research Assistant
Critical audit & apparatus collation (supervisor work)

→ Computer / AI
Initial candidate transcription (what R.A.s used to do)

R.A. AS SCHOLARLY AUDITOR

Demarquis Moss: Lead Auditor

- Ruled on disputed Latin readings against Chaparro Gómez's critical edition.
- Diagnostic Validity: Audited prompt suites for authentic palaeographical tests.
- Judge Calibrations: Screened AI judges against human scholarly ground truth.
- Editorial Leadership: Conducted supervisory research formerly reserved for postdocs.

→ Result: Every team member does higher-level work.

SLIDE 9 · PEDAGOGICAL FRAMEWORK

Teaching the Ladder: The Digital Methods Course

One phase a week, taught three ways: students must learn the entire ladder

MEETING 1: BY HAND

Direct Experience

- Type & decipher data manually.
- Builds palaeographical intuition.
- Foundation for critical literacy.

MEETING 2: PRE-AI TOOLS

Computational Tools

- Deterministic algorithms & regex.
- CollateX, treebanks, databases.
- Fast, rule-governed processing.

MEETING 3: GENERAL LLMs

Critical Editor

- Shape inputs: prompts & schemas.
- Curate & audit AI draft outputs.
- Trains real-world editorial judgment.

THE CARDINAL PEDAGOGICAL PRINCIPLE:

"You cannot judge a transcription you couldn't have made yourself."

SLIDE 10 · PRACTICAL RECOMMENDATIONS

Model Recommendations: This Week's Snapshot

September 2026 snapshot — models change rapidly, but workflow principles endure

PROJECT NEED	RECOMMENDED MODEL	BENCHMARK EVIDENCE	COST / SUITE
Budget Project	Gemini 3.7 Flash (low)	80.4 score · 87s latency	$0.14 / suite
Balanced Production	GPT-6 Astra (none)	83.5 score · 292s latency	$0.60 / suite
Highest Quality	Claude Fable 5.1 (none)	90.3 score · 504s latency	$1.72 / suite
Collation	Grok 4.7 (low) / GPT-6 Astra	96.7 / 94.5 task score	Sub-task
Translation	Gemini 3.1 Pro / Grok 4.7	97.0 / 97.0 task score	Sub-task

1. TURN REASONING DOWN

High effort costs 4× more, runs 5× slower, and scores lower on philology.

2. BEWARE FREE TIERS

Good for tests, but fail on production runs due to hard quota exhaustion.

3. RUN DUAL FAMILIES

Compare two distinct models; loci where they disagree pinpoint human review.

CONCLUSION & CORE THESIS

From Author to Editor: Elevating the Team

AI does not replace the scholar. It moves every member of the team up one rung: the machine drafts, the R.A. edits, and human effort concentrates at the start and end of the project.

OPEN RESEARCH DELIVERABLES

- editio-bench: Open 13-task benchmark suite & consensus scoring harness.
- PROVENANCE.md: Complete registry mapping every published score to its run.
- Clean Suite Master: 57 verified configurations passing quality & replay gates.
- GitHub Repository: github.com/kmpenner/IDG (XML ground truth, code, data).

PRESENTERS & SUPPORT

Dr. Ken M. Penner
Principal Investigator · St. Francis Xavier University
Email: kpenner@stfx.ca

Demarquis Moss
Lead Research Assistant & Scholarly Quality Auditor

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PANEL REFERENCE

Panel Q&A Cheat Sheet

Anticipated technical and methodological questions for the discussion period

- **Q1: Are models just reciting pre-trained printed editions?**
  ➔ No. Greek p.11 has no published diplomatic edition; Latin editions are normalized; Aramaic failed completely (80% CER).
- **Q2: What does this pipeline cost in actual research practice?**
  ➔ Budget setup was $0.14 for all 13 tasks (~1 cent/task). Frontier suite was $1.72 total. No supercomputers needed.
- **Q3: How solid is the 90 / 10 / 1 rule of thumb?**
  ➔ Project working impression on legible texts, verified by Greek CER 0.0365 (96% right). Breaks on broken leather.
- **Q4: Does this deskill undergraduate students and R.A.s?**
  ➔ No, it elevates them. The course keeps weekly 'by-hand' meetings: you cannot edit what you couldn't write yourself.
- **Q5: How are copyright and manuscript image rights governed?**
  ➔ Ingested under academic fair dealing at calibrated resolutions. Raw facsimiles remain under institutional ownership.